

Runtime item	Recorded behavior	Interpretation boundary
Sampling	21 application runs per member/device; manually uploaded real-world images	Image identity/order was not standardized across devices; device means are not a strict matched-image hardware ranking.
Platform	Android only	No iPhone/iOS test was performed because no compatible test device was available.

Table I.4. Observational branch-runtime summary

Device	All runs	No-seg branch	Seg-triggered runs	Seg-triggered mean	Overall mean
vivo Y100	21	14; 5.016 s mean	7	6.914 s	5.649 s
Samsung A32	21	2; 0.499 s mean	19	2.155 s	1.997 s
POCO M7 Pro 5G	21	13; 1.860 s mean	8	2.641 s	2.157 s

The Samsung A32 no-seg rows are two below-threshold short-circuit cases and are not a stable Healthy-branch hardware estimate. On vivo Y100, eight Healthy above-threshold runs averaged 5.024 s; on POCO M7 Pro 5G, ten Healthy above-threshold runs averaged 1.903 s. Overall device means depend strongly on branch composition and are therefore not used as a strict hardware ranking.

The retained operational photo set contains 21 usable real-world shrimp images collected across the three members' phone cameras, with variation in lighting, blur/softness, viewpoint, orientation, and background clutter. One unusably dark image was excluded. The set has no veterinary or laboratory ground truth and is an operational acquisition case study, not a field-accuracy benchmark.

Appendix J - Reproducibility and Artifact Checklist

This checklist records the current preservation status of research and deployment evidence. "Retained" means the evidence is present in the document/artifact set; it does not imply that every artifact has an independently verified cryptographic chain.

Table J.1. Re-audited reproducibility and artifact checklist

Artifact / control	Status	Evidence boundary or required action
Primary class mapping and counts	Retained	SDI-4 four-class order and fixed split counts are documented consistently.

Artifact / control	Status	Evidence boundary or required action
Fixed seed and split policy	Retained	Seed 42 and primary classification partition statistics are retained.
Classification training configuration	Retained	Optimizer, schedule, budget, augmentation family, and checkpoint logic are recorded.
Representative candidate ledger	Partially retained	Complete joint rows exist for five Ultralytics candidates; remaining TIMM/YOLO joint metrics are not reconstructed.
Primary classifier checkpoint / export identity	Partially retained	Checkpoint and export hashes exist, but the archived weight audit records a train-args naming conflict; independent architecture/evaluation verification remains required before using the candidate weight as sole proof of the final 91.01% Macro-F1 result.
Classifier FP32 TFLite metadata	Retained	Filename, input/output shapes, class order, byte size, and SHA-256 are preserved by the weight audit; the explicit hash is reproduced with the deployment contract in Appendix I.
Primary prediction-level classification file	Not retained	Required to regenerate classwise reports, paired errors, and confusion matrices from frozen predictions.
Primary classwise report	Not retained	Must be regenerated only from frozen prediction-level evidence.
Primary count and normalized confusion matrices	Not retained	Must not be reconstructed from aggregate metrics.
Classification corruption aggregates	Retained	Five aggregate corruption families are documented.
Complete per-severity corruption ledger	Partially retained	Qualitative severity panels exist, but complete machine-readable per-severity metrics were not retained.
Original segmentation annotation statistics	Retained	1,149 images, 416 specimen groups, 746 disease-labeled images, 403 Healthy negatives, and 1,031 masks are documented.
Expanded segmentation inventory	Retained	1,452-image inventory and 303 unmatched-name additions are documented with the mixed grouped/stratified validity boundary.
Independent expert mask review	Not retained	Multi-annotator review and inter-annotator agreement remain recommended.

Artifact / control	Status	Evidence boundary or required action
Attention protocol ledger	Retained	Placement, augmentation, epochs, initialization, and software versions are documented.
Attention parameter/timing ledger	Retained	Reference and four attention configurations retain parameter counts and notebook timing; values are not Android latency.
CA→SimAM placement study	Retained	Seven insertion patterns are documented; P3/P4/P5 is the retained final placement.
Strict matched broad attention ablation	Not retained	Broader attention-family rows use non-identical protocols; future causal comparison should equalize all non-attention controls.
Fourier candidate and timing ledgers	Retained	W1 parameters, candidate metrics, transform order, and CUDA timing scope are documented.
Paired W1 spectrum study	Retained	24 images × 12 matched realizations = 288 paired global observations, with retained high-band and gradient-effect summaries.
Final audited FDDEM checkpoint/result	Not retained	No numerical FDDEM claim should be made without a frozen checkpoint-result pair.
RTMDet-Ins-S validation/test evidence	Retained	Validation trajectory, epoch-86 checkpoint selection, and test comparator row are retained; cross-framework recall definitions remain non-identical.
Final segmentation mobile meta-data	Partially retained	YOLO11n-seg + CA→SimAM, FP16 filename, 5.65 MB size, decoder contract, and app use are documented; segmentation artifact hash is not retained here.
Three-device Android runtime logs	Retained	Three CSV logs with 21 runs each and the common application/model configuration are available; branch mixes differ.
Three-camera operational image set	Retained	21 usable photographs are retained as operational inputs; no veterinary/lab ground truth is available.
Multi-seed final result distribution	Not retained	Run three to five immutable-seed repetitions for uncertainty estimation.
Environment lockfile and dependency hash	Partially retained	Key framework versions are recorded for several runs; a unified lockfile and environment hash remain recommended.